BUSFIN 8200: Asset Pricing Foundations
Problem Set 2

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Question 1

1(a)

Yes.

[Definition of Complete Markets will be here]

Let

$$\mathbf{X} = \begin{bmatrix} 10 & 20 & 30 \\ 10 & 10 & 10 \\ 0 & 10 & 10 \end{bmatrix}$$

(rows: assets A, B, C; columns: states $n = 1, 2, 3$). Then there exists $\mathbf{\Omega}$ such that $\mathbf{X}\mathbf{\Omega} = \mathbf{I}$, with $\mathbf{\Omega} = \mathbf{X}^{-1}$, if $\mathbf{X}$ is invertible. Therefore, checking whether $\mathbf{X}$ is invertible is sufficient for Complete Markets in this economy. $\mathbf{X}$ has full rank. Therefore, this market is complete.

1(b)

Setup. Let $P_c(n)$ denote the price of the contingent claim for state n. Then

$$P(A) = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}_A(n),$$
$$P(B) = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}_B(n),$$
$$P(C) = \sum_{n=1}^{3} P_c(n)\,\text{Payoff}_C(n).$$

Method. Or, in matrix form,

$$\mathbf{P} = \mathbf{X}\mathbf{P}_c,$$

where $\mathbf{X}$ is the payoff matrix from Question 1(a),

$$\mathbf{P}_c = \big(P_c(1), P_c(2), P_c(3)\big)', \quad \text{and} \quad \mathbf{P} = \big(P(A), P(B), P(C)\big)' = (16, 8, 6)'.$$

Result. We confirmed in Question 1(a) that the payoff matrix is invertible. So, we can calculate

$$\mathbf{P}_c = \mathbf{X}^{-1}\mathbf{P} = \textit{[numerical values to be computed by the student]}.$$

1(c)

Method. As calculated in Question 1(b), $\mathbf{P}_c =$ *[values from 1b)]*. Taking the inner product of $\mathbf{Payoff}(D) = (30, 20, 10)'$ with $\mathbf{P}_c$ gives the price of asset D in this economy:

Result.

$$P(D) = \mathbf{Payoff}(D)' \, \mathbf{P}_c = \textit{[to be computed by the student]}.$$

1(d)

Setup. The SDF M satisfies $\mathbb{E}[M \cdot \text{Payoff}] = \text{Price}$ for every asset in this economy:

$$P(A) = \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_A(n),$$

$$P(B) = \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_B(n),$$

$$P(C) = \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_C(n).$$

Method. Or, in matrix form,

$$\mathbf{P} = \mathbf{X}_\pi \mathbf{M}, \qquad \mathbf{X}_\pi = \begin{bmatrix} 10 \cdot \frac{1}{4} & 20 \cdot \frac{1}{2} & 30 \cdot \frac{1}{4} \\ 10 \cdot \frac{1}{4} & 10 \cdot \frac{1}{2} & 10 \cdot \frac{1}{4} \\ 0 & 10 \cdot \frac{1}{2} & 10 \cdot \frac{1}{4} \end{bmatrix}$$

(rows: assets A, B, C; columns: states $n = 1, 2, 3$), where $\mathbf{M} = \big(M(1), M(2), M(3)\big)'$ and $\mathbf{P} = \big(P(A), P(B), P(C)\big)'$.

Result. $\mathbf{X}_\pi$ is invertible if $\det(\mathbf{X}_\pi) \neq 0$: $\det(\mathbf{X}_\pi) =$ *[to be computed by the student]*. Therefore, the SDF realizations are

$$\mathbf{M} = \mathbf{X}_\pi^{-1} \mathbf{P} = \textit{[numerical values to be computed by the student]}.$$

Interpretation. This economy satisfies No-Arbitrage.

[Definition of No-Arbitrage will be here]

As in Question 1(b), every contingent claim satisfies the No-Arbitrage condition above.

As we can see, the realized SDF is the same in every state of the economy. It may imply that each state is equally valuable *[?]*.

1(e)

Method. Using the SDF from Question 1(d), with $\text{Payoff}_D(1) = 30$, $\text{Payoff}_D(2) = 20$, and $\text{Payoff}_D(3) = 10$,

Result.

$$P(D) = \sum_{n=1}^{3} \pi(n)\, M(n)\, \text{Payoff}_D(n) = \textit{[to be computed by the student; compare with 1c)]}.$$

Question 2

2(a)

Setup. The SDF $M_t^* = a - \mathbf{b}'\mathbf{xR}_t$ prices the excess returns $\mathbf{xR}_t$ of the 10 value-weighted book-to-market decile portfolios ("Lo 10" to "Hi 10", Ken French data library), in excess of the Fama–French risk-free rate. Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months), in decimals.

Method. With the normalization $\widehat{\mathbb{E}}[M^*] = 1$, a and $\mathbf{b}$ solve the sample moment conditions

$$a - \mathbf{b}'\,\widehat{\mathbb{E}}[\mathbf{xR}] = 1,$$
$$a\,\widehat{\mathbb{E}}[\mathbf{xR}] - \widehat{\mathbb{E}}[\mathbf{xR}\,\mathbf{xR}']\,\mathbf{b} = \mathbf{0},$$

where $\widehat{\mathbb{E}}[\cdot]$ is the time average.

Result. Figure 1 plots $\hat{\mathbf{b}}$ by decile (estimates in `output/q2a_sdf_coefficients.csv`).

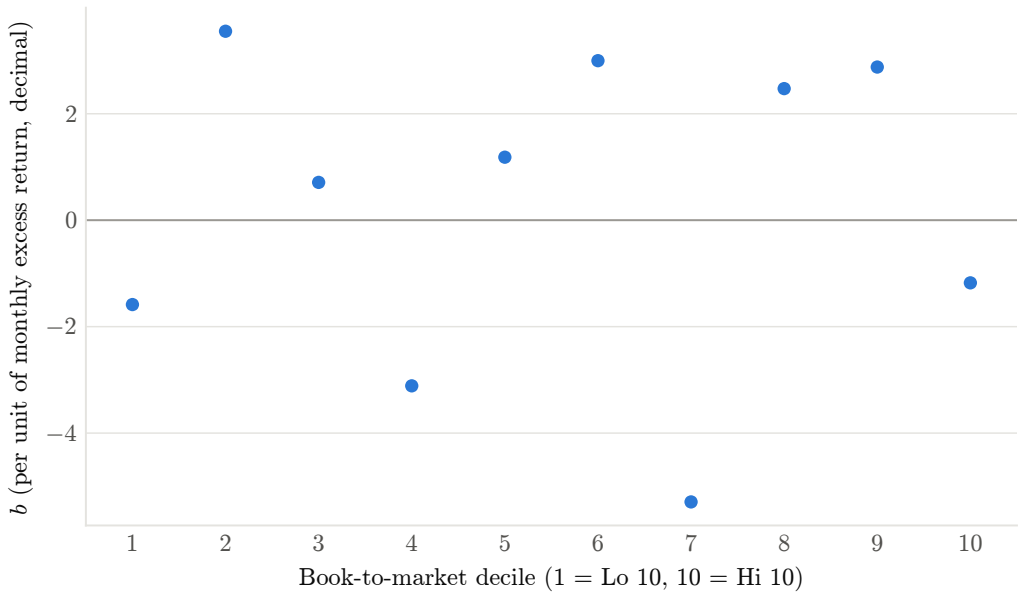


Figure 1: Coefficients $\hat{\mathbf{b}}$ of the SDF $M_t^* = a - \mathbf{b}'\mathbf{xR}_t$ that prices the excess returns of the 10 value-weighted book-to-market decile portfolios, under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months); excess returns in decimals, so b is per unit of decimal monthly excess return.

Interpretation. *[Describe the pattern you observe in the scatterplot.]*
[Explain why this pattern emerges.]

2(b)

Setup. As in Question 2(a), with the 10 value-weighted momentum decile portfolios ("Lo PRIOR" to "Hi PRIOR", prior 12–2 month return) as test assets. Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months), in decimals.

Method. The same moment conditions as in Question 2(a). The correlation of the two SDF series is the sample (Pearson) correlation.

Result. Figure 2 plots $\hat{\mathbf{b}}$ by decile. The sample (Pearson) correlation between the monthly SDF series $M^*_{\mathrm{BM},t}$ estimated in Question 2(a) and $M^*_{\mathrm{MOM},t}$ estimated in Question 2(b), over July 1927 to August 2026 ($T = 1190$ months), is

$$\widehat{\mathrm{Corr}}\left(M^*_{\mathrm{BM},t},\, M^*_{\mathrm{MOM},t}\right) = 0.355.$$

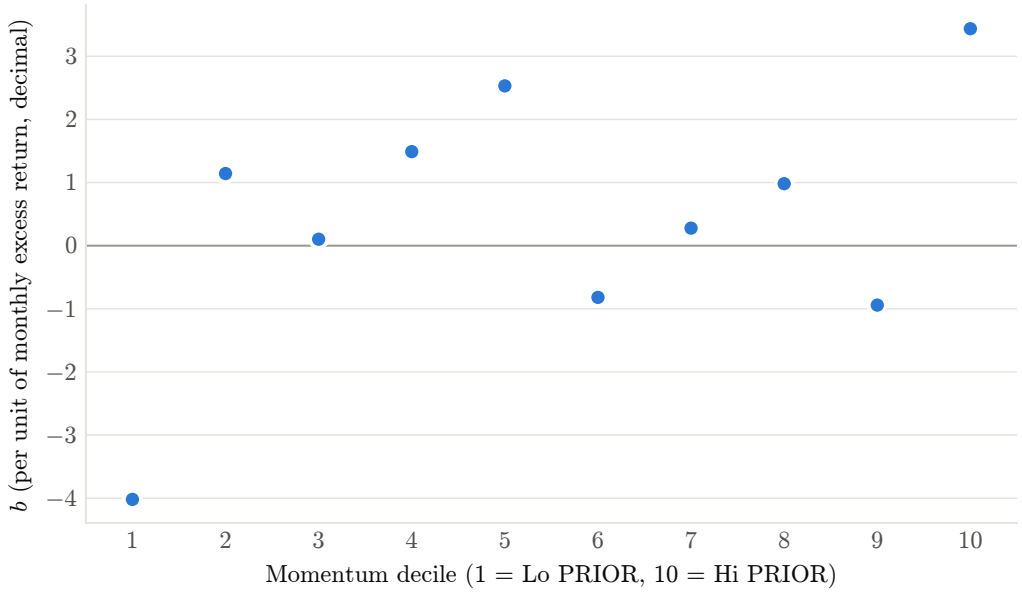


Figure 2: Coefficients $\hat{\mathbf{b}}$ of the SDF $M^*_t = a - \mathbf{b}'\mathbf{x}\mathbf{R}_t$ that prices the excess returns of the 10 value-weighted momentum (prior 12–2 month return) decile portfolios, under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months); excess returns in decimals, so b is per unit of decimal monthly excess return.

Interpretation. *[Describe the pattern you observe in the scatterplot.]*
[Explain why this pattern emerges.]
[Interpret this correlation in economic terms.]

2(c)

Setup. As in Question 2(a), with the 20 portfolios of Questions 2(a) and 2(b) together as test assets. Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months), in decimals.

Method. The same moment conditions as in Question 2(a). For the SDFs of Questions 2(a)–2(c), the Sharpe ratio of $xR_t^* = \hat{\mathbf{b}}'\mathbf{x}\mathbf{R}_t$ is $\widehat{SR}(xR^*) = \widehat{\mathbb{E}}[xR_t^*]/\widehat{\sigma}(xR_t^*)$, with $\widehat{\sigma}$ dividing by T.

Result. Figure 3 plots $\hat{\mathbf{b}}$ by decile and Table 1 reports the Sharpe ratios.

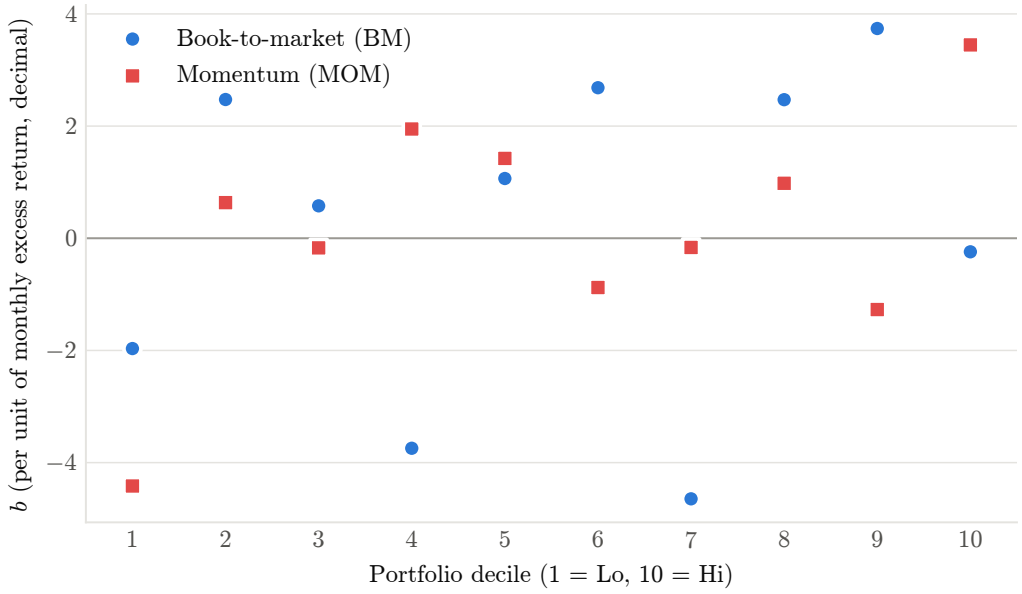


Figure 3: Coefficients $\hat{\mathbf{b}}$ of the SDF $M_t^* = a - \mathbf{b}'\mathbf{x}\mathbf{R}_t$ that prices the excess returns of the 20 value-weighted book-to-market (blue circles) and momentum (red squares) decile portfolios, under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months); excess returns in decimals, so b is per unit of decimal monthly excess return.

SDF	Test assets	$\widehat{SR}(xR^*)$
Question 2(a)	10 book-to-market deciles	0.192
Question 2(b)	10 momentum deciles	0.258
Question 2(c)	20 book-to-market and momentum deciles	0.305

Table 1: Sharpe ratios $\widehat{SR}(xR^*) = \widehat{\mathbb{E}}[xR_t^*]/\widehat{\sigma}(xR_t^*)$ of the portfolios $xR_t^* = \hat{\mathbf{b}}'\mathbf{x}\mathbf{R}_t$ implied by the SDFs of Questions 2(a)–2(c). Sample: July 1927 to August 2026 ($T = 1190$ months). Data: monthly returns (one-month horizon). Units: monthly Sharpe ratio, not annualized; $\widehat{\sigma}$ divides by T. No standard errors are reported.

Interpretation. *[Describe the pattern you observe in the scatterplot.]*

[Explain why this pattern emerges.]

[Explain what you observe in the Sharpe ratios (Table 1) and why it is (or it is not) in line with a key SDF property from the lectures.]

2(d)

Setup. Pricing errors of the 20 portfolios of Question 2(c) under each of the three SDFs M_t^* estimated in Questions 2(a), 2(b) and 2(c). Data: monthly returns (one-month horizon), July 1927 to August 2026 ($T = 1190$ months).

Method. For each portfolio p and each SDF,

$$\hat{\alpha}_p = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*, xR_p],$$
$$\widehat{\mathrm{Cov}}[-M^*, xR_p] = \frac{1}{T}\sum_{t=1}^{T}(-M_t^* + \overline{M^*})(xR_{p,t} - \overline{xR}_p).$$

Result. Table 2.

	SDF M_t^* estimated in		
Portfolio	2(a) BM	2(b) MOM	2(c) BM + MOM
Panel A: book-to-market deciles			
1 (Lo 10)	0.000	−0.151	0.000
2	0.000	−0.010	0.000
3	0.000	−0.022	0.000
4	0.000	−0.078	0.000
5	0.000	0.050	0.000
6	0.000	0.126	0.000
7	0.000	0.028	0.000
8	0.000	0.238	0.000
9	0.000	0.345	0.000
10 (Hi 10)	0.000	0.320	0.000
Panel B: momentum deciles			
1 (Lo PRIOR)	−1.016	0.000	0.000
2	−0.505	0.000	0.000
3	−0.327	0.000	0.000
4	−0.149	0.000	0.000
5	−0.088	0.000	0.000
6	−0.026	0.000	0.000
7	0.051	0.000	0.000
8	0.190	0.000	0.000
9	0.224	0.000	0.000
10 (Hi PRIOR)	0.542	0.000	0.000

Table 2: Pricing errors $\hat{\alpha}_p = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*, xR_p]$ of the 20 book-to-market and momentum decile portfolios under the SDFs M_t^* estimated in Question 2(a) (10 book-to-market deciles), Question 2(b) (10 momentum deciles) and Question 2(c) (all 20 portfolios). Sample: July 1927 to August 2026 ($T = 1190$ months). Data: monthly value-weighted returns (one-month horizon) in excess of the Fama–French risk-free rate. Units: percent per month. Covariances divide by T. No standard errors are reported.

Interpretation. *[Explanation will be here]*

2(e)

Setup. Out-of-sample pricing errors of the 20 portfolios of Question 2(c). Data: monthly returns (one-month horizon); estimation sample July 1927 to December 1989 ($T = 750$ months), evaluation sample January 1990 to August 2026 ($T = 440$ months).

Method. $\hat{a}$ and $\hat{\mathbf{b}}$ are estimated as in Question 2(c) on the estimation sample. The out-of-sample SDF is $M^*_{OS,t} = \hat{a} - \hat{\mathbf{b}}'\mathbf{x}\mathbf{R}_t$ on the evaluation sample, and $\hat{\alpha}_{OS,p} = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*_{OS}, xR_p]$, with all moments computed on the evaluation sample and covariances dividing by T.

Result. Table 3.

Portfolio	$\hat{\alpha}_{OS,p}$
Panel A: book-to-market deciles	
1 (Lo 10)	1.097
2	0.786
3	0.900
4	0.696
5	0.799
6	0.718
7	0.698
8	0.592
9	0.827
10 (Hi 10)	0.971
Panel B: momentum deciles	
1 (Lo PRIOR)	2.705
2	1.648
3	1.390
4	1.208
5	0.918
6	0.830
7	0.760
8	0.644
9	0.432
10 (Hi PRIOR)	0.545

Table 3: Out-of-sample pricing errors $\hat{\alpha}_{OS,p} = \widehat{\mathbb{E}}[xR_p] - \widehat{\mathrm{Cov}}[-M^*_{OS}, xR_p]$ of the 20 book-to-market and momentum decile portfolios. $\hat{a}$ and $\hat{\mathbf{b}}$ are estimated as in Question 2(c) on July 1927 to December 1989 ($T = 750$ months); $M^*_{OS,t} = \hat{a} - \hat{\mathbf{b}}'\mathbf{x}\mathbf{R}_t$ and the moments are computed on January 1990 to August 2026 ($T = 440$ months). Data: monthly value-weighted returns (one-month horizon) in excess of the Fama–French risk-free rate. Units: percent per month. Covariances divide by T. No standard errors are reported.

Interpretation. *[Explanation will be here]*

Question 3

3(a)

Method. From the maximization problem in Equation (3.2),

$$\boldsymbol{\varpi}_t^* = \arg\max_{\boldsymbol{\varpi}_t} \mathbb{E}_t\big[U(W_{t+1})\big] \quad \text{s.t.} \quad W_{t+1} = W_t \cdot \big(R_{f,t+1} + \boldsymbol{\varpi}_t' \mathbf{x}\mathbf{R}_{t+1}\big),$$

where $R_{w,t+1} = R_{f,t+1} + \boldsymbol{\varpi}_t^{*\prime} \mathbf{x}\mathbf{R}_{t+1}$ is the return on the optimal portfolio, so that $W_{t+1} = W_t \cdot R_{w,t+1}$.

The first-order condition is

$$\begin{aligned} & \frac{\partial\, \mathbb{E}_t[U(W_{t+1})]}{\partial \boldsymbol{\varpi}_t} = \mathbb{E}_t\big[U'(W_{t+1}) \cdot W_t \cdot \mathbf{x}\mathbf{R}_{t+1}\big] = \mathbf{0} \\ \iff \quad & \mathbb{E}_t\big[U'(W_{t+1}) \cdot \mathbf{x}\mathbf{R}_{t+1}\big] = \mathbf{0}. \end{aligned}$$

Since W_t is the investor's wealth at time t, it is positive ($W_t > 0$) and known at time t, so it can be taken outside $\mathbb{E}_t[\cdot]$ and divided out.

Because $U'(W_{t+1}) = W_{t+1}^{-\gamma} = (W_t \cdot R_{w,t+1})^{-\gamma}$,

$$\begin{aligned} & \mathbb{E}_t\big[(W_t \cdot R_{w,t+1})^{-\gamma} \cdot \mathbf{x}\mathbf{R}_{t+1}\big] = \mathbf{0} \\ \iff \quad & W_t^{-\gamma}\, \mathbb{E}_t\big[R_{w,t+1}^{-\gamma} \cdot \mathbf{x}\mathbf{R}_{t+1}\big] = \mathbf{0} \\ \iff \quad & \mathbb{E}_t\big[R_{w,t+1}^{-\gamma} \cdot \mathbf{x}\mathbf{R}_{t+1}\big] = \mathbf{0}, \end{aligned}$$

where the last step uses the same argument for $W_t^{-\gamma} > 0$.

Result. Therefore, $M_{t+1} = R_{w,t+1}^{-\gamma}$ prices the vector of excess returns. Because an SDF that prices only excess returns still does so after multiplication by any positive factor (here, $W_t^{-\gamma}$), the scale of M is not identified; a normalization is therefore needed and is innocuous.

3(b)

Setup. GMM estimate of γ from the unconditional SDF equation $\mathbb{E}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$, with $R_{w,t} = e^{r_{e,t}}$ and $xR_{w,t} = e^{r_{e,t}} - e^{r_{f,t}}$ from the EQ Dataset (annual returns observed monthly).

Method. With one moment and one parameter, $\hat{\gamma}$ solves $\hat{G} \cdot \widehat{\mathbb{E}}[R_{w,t}^{-\hat{\gamma}} \cdot xR_{w,t}] = 0$ with the analytical $\hat{G} = \widehat{\mathbb{E}}[\partial_\gamma(R_{w,t}^{-\gamma} \cdot xR_{w,t})]$ (footnote 5), and $\mathrm{aVar}[\hat{\gamma}] = \hat{S}/(T\hat{G}^2)$ with $\hat{S}$ the Newey–West estimate.

Result. Table 4.

	GMM
$\hat{\gamma}$	1.8873
	(0.5930)
Newey–West lag	23
Sample	Jun 1928 – Dec 2021
T (months)	1123

Table 4: GMM estimate of γ from $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$, with $R_{w,t} = e^{r_{e,t}}$ and $xR_{w,t} = e^{r_{e,t}} - e^{r_{f,t}}$ from the EQ Dataset. Sample: June 1928 to December 2021 ($T = 1123$ months). Data: monthly observations of 12-month (annual) real returns. Units: decimals. Standard error in parentheses below the estimate: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down (selected lag in the table).

3(c)

Setup. GMM estimate of γ from the two moment conditions $\mathbb{E}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$ and $\mathbb{E}[R_{w,t}^{-\gamma} \cdot (z_{t-1} \cdot xR_{w,t})] = 0$, with $R_{w,t}$ and $xR_{w,t}$ as in Question 3(b). The instrument is $z_{t-1} = \widehat{\mathbb{E}}^{OS}[xR_e]/\sigma_e^2$, where $\widehat{\mathbb{E}}^{OS}[xR_e]$ is the Problem Set 1 (2e) out-of-sample forecast made twelve months before t (the forecast of $xR_{w,t}$) and σ_e^2 is the variance of $R_{e,t}$ over the sample (dividing by $T - 1$). Sample: December 1940 to December 2021.

Method. With $\hat{\mathbf{W}} = \mathbf{I}$, so $\hat{\mathbf{A}} = \hat{\mathbf{G}}'$, $\hat{\gamma}$ solves $\hat{\mathbf{G}}' \widehat{\mathbb{E}}[\boldsymbol{\epsilon}(\hat{\gamma})] = 0$, where $\widehat{\mathbb{E}}[\boldsymbol{\epsilon}(\gamma)]$ is the 2×1 vector of sample moments and $\hat{\mathbf{G}}$ its analytical derivative. Then

$$\begin{aligned} \operatorname{aVar}[\hat{\gamma}] &= \tfrac{1}{T}(\hat{\mathbf{A}}\hat{\mathbf{G}})^{-1}\hat{\mathbf{A}}\hat{\mathbf{S}}\hat{\mathbf{A}}'(\hat{\mathbf{A}}\hat{\mathbf{G}})^{-1\prime}, \\ \operatorname{aVar}\big[\widehat{\mathbb{E}}[\boldsymbol{\epsilon}]\big] &= \tfrac{1}{T}\big[\mathbf{I} - \hat{\mathbf{G}}(\hat{\mathbf{A}}\hat{\mathbf{G}})^{-1}\hat{\mathbf{A}}\big]\hat{\mathbf{S}}\big[\mathbf{I} - \hat{\mathbf{G}}(\hat{\mathbf{A}}\hat{\mathbf{G}})^{-1}\hat{\mathbf{A}}\big]', \end{aligned}$$

with $\hat{\mathbf{S}}$ the Newey–West estimate. Each pricing error has $t_j = \widehat{\mathbb{E}}[\epsilon]_j / \sqrt{\operatorname{aVar}_{jj}}$; the joint test is $\widehat{\mathbb{E}}[\boldsymbol{\epsilon}]' \operatorname{aVar}^{+} \widehat{\mathbb{E}}[\boldsymbol{\epsilon}] \sim \chi^2(1)$, with the Moore–Penrose pseudo-inverse of the rank-one covariance matrix.

Result. Table 5.

	Estimate	s.e.
$\hat{\gamma}$	3.0130	0.7156
Pricing error	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
Market, $xR_{w,t}$	-0.0040	-0.86
Managed portfolio, $z_{t-1} \cdot xR_{w,t}$	0.0019	0.86
$\chi^2(1)$	0.75	
p-value	0.388	
Newey–West lag	21	
Sample	Dec 1940 – Dec 2021 ($T = 973$)	

Table 5: GMM estimate of γ from $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot xR_{w,t}] = 0$ and $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot (z_{t-1} \cdot xR_{w,t})] = 0$ with an identity weighting matrix ($\hat{\mathbf{A}} = \hat{\mathbf{G}}'$); z_{t-1} as defined in the text. Sample: December 1940 to December 2021 ($T = 973$ months). Data: monthly observations of 12-month (annual) real returns from the EQ Dataset. Units: decimals. Standard error next to the estimate and t-statistics: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down (selected lag in the table). χ^2: test that the two pricing errors are jointly zero (Moore–Penrose pseudo-inverse).

3(d)

Setup. The moment conditions, data, instrument and sample of Question 3(c), estimated by two-step GMM.

Method. Step 1 is the estimate of Question 3(c) ($\hat{\mathbf{W}} = \mathbf{I}$), and $\hat{\mathbf{S}}_1$ is the Newey–West estimate at that estimate. Step 2 uses $\hat{\mathbf{W}} = \hat{\mathbf{S}}_1^{-1}$: $\hat{\gamma}$ solves $\hat{\mathbf{G}}'\hat{\mathbf{S}}_1^{-1}\,\widehat{\mathbb{E}}[\boldsymbol{\epsilon}(\hat{\gamma})] = 0$, with no further iteration. Standard error, t-statistics and χ^2 use the formulas of Question 3(c) with $\hat{\mathbf{A}} = \hat{\mathbf{G}}'\hat{\mathbf{S}}_1^{-1}$, $\hat{\mathbf{G}}$ at the step-2 estimate, and $\hat{\mathbf{S}}_2$, the Newey–West estimate re-computed at the step-2 estimate.

Result. Table 6.

	Estimate	s.e.
$\hat{\gamma}$ (two-step)	2.9559	0.7058
Pricing error	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
Market, $xR_{w,t}$	-0.0022	-0.88
Managed portfolio, $z_{t-1} \cdot xR_{w,t}$	0.0058	0.88
$\chi^2(1)$	0.77	
p-value	0.379	
Newey–West lag, $\hat{S}_1$ / $\hat{S}_2$	21 / 21	
Sample	Dec 1940 – Dec 2021 ($T = 973$)	

Table 6: Two-step GMM estimate of γ from the moment conditions of Question 3(c); $\hat{\mathbf{W}} = \hat{\mathbf{S}}_1^{-1}$ in step 2 and $\hat{\mathbf{A}} = \hat{\mathbf{G}}'\hat{\mathbf{S}}_1^{-1}$ in the sandwich. Sample: December 1940 to December 2021 ($T = 973$ months). Data: monthly observations of 12-month (annual) real returns from the EQ Dataset. Units: decimals. Standard error next to the estimate and t-statistics: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down, re-selected for $\hat{\mathbf{S}}_1$ and $\hat{\mathbf{S}}_2$ (selected lags in the table). χ^2: test that the two pricing errors are jointly zero (Moore–Penrose pseudo-inverse).

3(e)

Setup. GMM estimate of γ from $\mathbb{E}[R_{w,t}^{-\gamma} \cdot xR_{j,t}] = 0$ for $j = 1, \ldots, 21$: the trailing 12-month excess returns of the 20 value-weighted book-to-market and momentum decile portfolios (nominal, over the 12-month compounded Fama–French risk-free rate) and $xR_{w,t}$. Sample: December 1940 to December 2021.

Method. As in Question 3(c) with 21 moments: $\hat{\mathbf{W}} = \mathbf{I}$, $\hat{\mathbf{A}} = \hat{\mathbf{G}}'$, the same variance formulas, and the joint test $\widehat{\mathbb{E}}[\boldsymbol{\epsilon}]' \, \mathrm{aVar}^{+} \, \widehat{\mathbb{E}}[\boldsymbol{\epsilon}] \sim \chi^2(20)$ (rank 20).

Result. Table 7.

	Estimate	s.e.
$\hat{\gamma}$	2.9920	0.7020
Pricing error	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
Book-to-market deciles		
1 (Lo 10)	−0.0173	−1.52
2	0.0014	0.21
3	−0.0038	−0.65
4	0.0005	0.08
5	0.0151	2.32
6	0.0205	2.67
7	0.0058	0.49
8	0.0299	3.02
9	0.0401	3.89
10 (Hi 10)	0.0335	2.17
Momentum deciles		
1 (Lo PRIOR)	−0.1048	−7.70
2	−0.0436	−4.92
3	−0.0199	−2.30
4	−0.0025	−0.37
5	0.0013	0.18
6	0.0019	0.34
7	0.0092	1.23
8	0.0199	3.84
9	0.0227	3.18
10 (Hi PRIOR)	0.0500	4.32
Market		
$xR_{w,t}$	−0.0033	−0.78
$\chi^2(20)$	197.88	
p-value	<0.001	
Newey–West lag	21	
Sample	Dec 1940 – Dec 2021 ($T = 973$)	

Table 7: GMM estimate of γ from $\widehat{\mathbb{E}}[R_{w,t}^{-\gamma} \cdot xR_{j,t}] = 0$, $j = 1, \ldots, 21$, with an identity weighting matrix ($\hat{\mathbf{A}} = \hat{\mathbf{G}}'$). Sample: December 1940 to December 2021 ($T = 973$ months). Data: monthly observations of 12-month returns: the 20 decile portfolios (value-weighted, nominal, over the 12-month compounded risk-free rate) and $xR_{w,t} = e^{r_{e,t}} - e^{r_{f,t}}$ (EQ Dataset, real); $R_{w,t} = e^{r_{e,t}}$. Units: decimals. Standard error next to the estimate and t-statistics: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down (selected lag in the table). χ^2: test that the 21 pricing errors are jointly zero (Moore–Penrose pseudo-inverse).

Question 4

4(a)

Conditionally linear CAPM. Let $M_{t+1} = M(F_{t+1};\theta)$. It can be approximated by a Taylor expansion around $\mathbb{E}_t[F_{t+1}]$:

$$M(F_{t+1};\theta) = \sum_{n=0}^{\infty} \frac{M^{(n)}\big(\mathbb{E}_t[F_{t+1}];\theta\big)}{n!}\big(F_{t+1} - \mathbb{E}_t[F_{t+1}]\big)^n.$$

Terms of second order and above can be neglected. Then,

$$M(F_{t+1};\theta) \approx M\big(\mathbb{E}_t[F_{t+1}];\theta\big) + M'\big(\mathbb{E}_t[F_{t+1}];\theta\big)\big(F_{t+1} - \mathbb{E}_t[F_{t+1}]\big).$$

If we plug in $M(F_{t+1};\theta) = F_{t+1}^{-\gamma}$ with $F_{t+1} = R_{w,t+1}$, then

$$M_{t+1} \approx -\gamma\,\mathbb{E}_t[R_{w,t+1}]^{-\gamma-1}R_{w,t+1} + (\gamma+1)\,\mathbb{E}_t[R_{w,t+1}]^{-\gamma}.$$

Therefore,

$$a_t = (\gamma+1)\,\mathbb{E}_t[R_{w,t+1}]^{-\gamma}, \qquad b_t = \gamma\,\mathbb{E}_t[R_{w,t+1}]^{-\gamma-1}.$$

Disadvantage relative to $M_{t+1} = R_{w,t+1}^{-\gamma}$.

1. The original model, $M_{t+1} = R_{w,t+1}^{-\gamma}$, contains only the factor and the parameter, so there is no concern about additional assumptions or time-varying coefficients.

2. After linearization, a_t and b_t become time-varying, so $\mathbb{E}[xR_j]$ can no longer be written in the form $\mathbb{E}[b_t]'\,\mathrm{Cov}(F_t, xR_{j,t})$.

3. The time variation in a_t and b_t comes from $\mathbb{E}_t[R_{w,t+1}]$, as shown above. This conditional expectation must be estimated, and the choice of how to estimate it (e.g., with a linear model or some other method) introduces estimation error. The model could be wrong, or there could be estimation error even if the model is correct.

4. By imposing additional assumptions or a model on a_t and b_t, they can be made time-invariant. However, if that model is wrong or the assumptions are inconsistent with reality, the estimates move further away from reality.

5. $\mathbb{E}_t[\mathbf{xR}_{t+1}] = b_t'\,\mathrm{Cov}_t(F_{t+1}, \mathbf{xR}_{t+1})$ holds, but from the perspective of the unconditional implications used in GMM, it can be written as $\mathbb{E}[xR_j] = \mathbb{E}\big[b_t'\,\mathrm{Cov}_t(F_{t+1}, xR_{j,t+1})\big]$.

6. But again, b_t is determined by $\mathbb{E}_t[R_{w,t}]$. As in Problem Set 1, we already checked that estimating the equity premium *very well*, either in sample (IS) or out of sample (OS), is really difficult.

4(b)

Setup. SDF $M_t = 1 - b\,(R_{w,t} - \mu)$ with $\mu = \mathbb{E}[R_{w,t}]$ (normalization $\mathbb{E}[M] = 1$). Test assets: the 21 excess returns of Question 3(e) ($xR_{j,t}$, $j = 1,\ldots,20$, and $xR_{w,t}$), with $R_{w,t} = e^{r_{e,t}}$. Sample: June 1928 to December 2021.

Method. The parameters $\boldsymbol{\theta} = (b, \mu)$ satisfy 22 moments: the pricing moments $\mathbb{E}[M_t \cdot xR_{j,t}] = 0$, $j = 1, \ldots, 21$, and $\mathbb{E}[R_{w,t} - \mu] = 0$. The last one is set exactly to zero, so $\hat{\mu} = \overline{R}_w$, and b uses identity weights on the pricing moments:

$$\hat{b} = (\hat{\mathbf{d}}'\hat{\mathbf{d}})^{-1}\hat{\mathbf{d}}'\,\widehat{\mathbb{E}}[\mathbf{xR}], \qquad \hat{\mathbf{d}} = \widehat{\mathrm{Cov}}[R_{w,t}, \mathbf{xR}_t] \text{ (dividing by } T).$$

This is GMM with $\hat{\mathbf{A}} = \begin{bmatrix} \hat{\mathbf{G}}_b' & 0 \\ \mathbf{0}' & 1 \end{bmatrix}$, where $\hat{\mathbf{G}}_b$ is the derivative of the pricing moments with respect to b (rows: parameters b and μ; columns: the 21 pricing moments and the μ moment). Standard errors, t-statistics and the joint test use the formulas of Question 3(c) with this $\hat{\mathbf{A}}$, the 22×22 Newey–West $\hat{\mathbf{S}}$, and the 21×21 pricing block of $\mathrm{aVar}\left[\widehat{\mathbb{E}}[\boldsymbol{\epsilon}]\right]$ (rank 20), so they account for estimating μ.

Result. Table 8.

	Estimate	s.e.
$\hat{b}$	1.9751	0.6047
	$\widehat{\mathbb{E}}[\epsilon_j]$	t-stat
Book-to-market deciles		
1 (Lo 10)	−0.0062	−0.56
2	0.0163	1.98
3	0.0054	0.80
4	−0.0017	−0.28
5	0.0109	1.60
6	0.0163	2.12
7	−0.0014	−0.15
8	0.0195	2.34
9	0.0266	2.56
10 (Hi 10)	0.0083	0.56
Momentum deciles		
1 (Lo PRIOR)	−0.1110	−8.76
2	−0.0493	−5.45
3	−0.0275	−3.00
4	−0.0128	−1.84
5	0.0000	0.00
6	0.0003	0.06
7	0.0163	2.16
8	0.0298	4.57
9	0.0360	4.02
10 (Hi PRIOR)	0.0709	5.10
Market		
$xR_{w,t}$	−0.0002	−0.03
$\chi^2(20)$	229.37	
p-value	<0.001	
Newey–West lag	22	
Sample	Jun 1928 – Dec 2021 ($T = 1123$)	

Table 8: First-stage GMM estimate of b in $M_t = 1 - b\,(R_{w,t} - \mu)$ with identity weights on the 21 pricing moments and $\mathbb{E}[R_{w,t} - \mu] = 0$ satisfied exactly. Sample: June 1928 to December 2021 ($T = 1123$ months). Data: monthly observations of 12-month returns: the 20 decile portfolios (value-weighted, nominal, over the 12-month compounded risk-free rate) and $xR_{w,t} = e^{r_{e,t}} - e^{r_{f,t}}$ (EQ Dataset, real); $R_{w,t} = e^{r_{e,t}}$. Units: decimals. Standard error next to the estimate and t-statistics: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down (selected lag in the table). χ^2: test that the 21 pricing errors are jointly zero (Moore–Penrose pseudo-inverse).

4(c)

Setup. The model, data and sample of Question 4(b), estimated by efficient GMM.

Method. With $\hat{\mu} = \overline{R}_w$, for any weighting matrix $\hat{\mathbf{W}}$ on the 21 pricing moments b has the closed form

$$\hat{b}(\hat{\mathbf{W}}) = (\hat{\mathbf{d}}'\hat{\mathbf{W}}\hat{\mathbf{d}})^{-1}\hat{\mathbf{d}}'\hat{\mathbf{W}}\,\widehat{\mathbb{E}}[\mathbf{xR}],$$

so no numerical optimization is needed. Two-step: $\hat{\mathbf{W}} = \hat{\mathbf{S}}^{-1}$, with $\hat{\mathbf{S}}$ the 21×21 pricing block of the Newey–West estimate at the estimate of Question 4(b). Iterated: $\hat{\mathbf{S}}$ and $\hat{b}$ are updated until $|\hat{b}_{h+1} - \hat{b}_h| < 10^{-10}$. Standard errors, t-statistics and χ^2 use the formulas of Question 4(b) with $\hat{\mathbf{A}} = \begin{bmatrix} \hat{\mathbf{G}}_b'\hat{\mathbf{W}} & 0 \\ \mathbf{0}' & 1 \end{bmatrix}$ and $\hat{\mathbf{S}}$ re-estimated at each estimate.

Result. Table 9.

	4(b), $\hat{\mathbf{W}} = \mathbf{I}$		Two-step		Iterated	
$\hat{b}$	1.9751 (0.6047)		3.9531 (0.6284)		4.1161 (0.6231)	
Pricing error	$\widehat{\mathbb{E}}[\epsilon_j]$	t	$\widehat{\mathbb{E}}[\epsilon_j]$	t	$\widehat{\mathbb{E}}[\epsilon_j]$	t
Book-to-market deciles						
1 (Lo 10)	−0.0062	−0.56	−0.0926	−3.41	−0.0998	−3.23
2	0.0163	1.98	−0.0614	−2.99	−0.0678	−3.07
3	0.0054	0.80	−0.0751	−3.44	−0.0817	−2.98
4	−0.0017	−0.28	−0.0932	−2.80	−0.1008	−2.41
5	0.0109	1.60	−0.0750	−2.38	−0.0821	−2.17
6	0.0163	2.12	−0.0744	−2.08	−0.0818	−1.90
7	−0.0014	−0.15	−0.0959	−2.46	−0.1037	−2.19
8	0.0195	2.34	−0.0847	−2.28	−0.0933	−1.96
9	0.0266	2.56	−0.0846	−2.20	−0.0937	−1.83
10 (Hi 10)	0.0083	0.56	−0.1212	−2.54	−0.1319	−2.10
Momentum deciles						
1 (Lo PRIOR)	−0.1110	−8.76	−0.2343	−5.57	−0.2445	−4.75
2	−0.0493	−5.45	−0.1581	−3.85	−0.1670	−3.25
3	−0.0275	−3.00	−0.1185	−3.41	−0.1260	−3.13
4	−0.0128	−1.84	−0.1069	−2.91	−0.1146	−2.54
5	0.0000	0.00	−0.0838	−2.89	−0.0907	−2.51
6	0.0003	0.06	−0.0883	−2.81	−0.0956	−2.47
7	0.0163	2.16	−0.0640	−2.51	−0.0706	−2.43
8	0.0298	4.57	−0.0531	−2.28	−0.0599	−2.14
9	0.0360	4.02	−0.0502	−2.14	−0.0573	−2.09
10 (Hi PRIOR)	0.0709	5.10	−0.0210	−0.90	−0.0286	−1.16
Market						
$xR_{w,t}$	−0.0002	−0.03	−0.0833	−3.10	−0.0901	−2.86
$\chi^2(20)$	229.37		132.44		128.64	
p-value	<0.001		<0.001		<0.001	
NW lag, $\hat{\mathbf{W}}$ / sandwich	– / 22		22 / 20		20 / 20	
Iterations	–		–		32	
Sample	Jun 1928 – Dec 2021 ($T = 1123$)					

Table 9: Efficient GMM estimates of b in $M_t = 1 - b\,(R_{w,t} - \mu)$: Question 4(b) ($\hat{\mathbf{W}} = \mathbf{I}$, Table 8, for reference), two-step, and iterated. Sample: June 1928 to December 2021 ($T = 1123$ months). Data: monthly observations of 12-month returns, as in Table 8. Units: decimals. Standard errors in parentheses below the estimates and t-statistics: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down and re-selected each time $\hat{\mathbf{S}}$ is estimated (lags of the weighting matrix and of the sandwich in the table). χ^2: test that the 21 pricing errors are jointly zero (Moore–Penrose pseudo-inverse). "–": not applicable.

4(d)

Setup. Time-series estimates of the linear beta pricing model implied by $M_t = a - b \cdot R_{w,t}$. With $R_{f,t}$ treated as constant, $M_t = (a - bR_f) - b \cdot xR_{w,t}$, so the factor is the excess return $F_t = xR_{w,t}$. Test assets and sample as in Question 4(b) (June 1928 to December 2021).

Method. $\hat{\lambda} = \widehat{\mathbb{E}}[F]$, with the Newey–West standard error from a time-series regression of F_t on a constant. For each asset, $\hat{\alpha}_j$ is the intercept of the time-series regression $xR_{j,t} = \alpha_j + \beta_j F_t + \varepsilon_{j,t}$, with a Newey–West t-statistic. The GRS test (Gibbons, Ross and Shanken, 1989) for the 20 decile portfolios is

$$\text{GRS} = \frac{T - J - K}{J}\left(1 + \bar{F}'\hat{\Sigma}_F^{-1}\bar{F}\right)^{-1}\hat{\boldsymbol{\alpha}}'\hat{\boldsymbol{\Sigma}}_\varepsilon^{-1}\hat{\boldsymbol{\alpha}} \sim F(J, T - J - K),$$

with $J = 20$, $K = 1$, $\hat{\boldsymbol{\Sigma}}_\varepsilon$ dividing by $T - 2$ and $\hat{\Sigma}_F$ by $T - 1$.

Result. Table 10.

	Estimate	s.e.
$\hat{\lambda}$	0.0828	0.0186
	$\hat{\alpha}_j$	t-stat
Book-to-market deciles		
1 (Lo 10)	−0.0052	−0.64
2	0.0171	2.90
3	0.0064	0.97
4	−0.0017	−0.21
5	0.0104	1.20
6	0.0158	1.60
7	−0.0018	−0.15
8	0.0193	1.57
9	0.0272	1.76
10 (Hi 10)	0.0086	0.42
Momentum deciles		
1 (Lo PRIOR)	−0.1121	−7.55
2	−0.0487	−4.09
3	−0.0279	−3.25
4	−0.0129	−1.42
5	0.0003	0.04
6	0.0007	0.13
7	0.0161	2.59
8	0.0298	4.82
9	0.0361	4.20
10 (Hi PRIOR)	0.0712	5.51
Market (factor)		
$xR_{w,t}$	0.0000	–
GRS, $F(20, 1102)$	49.65	
p-value	<0.001	
Sample	Jun 1928 – Dec 2021 ($T = 1123$)	

Table 10: Time-series estimates of λ and α_j with the factor $F_t = xR_{w,t}$. Sample: June 1928 to December 2021 ($T = 1123$ months). Data: monthly observations of 12-month returns, as in Table 8. Units: decimals. Standard error of $\hat{\lambda}$ next to the estimate and t-statistics of $\hat{\alpha}_j$: Newey–West (1987, 1994), Bartlett kernel, Newey–West (1994) automatic lag rounded down, selected for each regression (lag 23 for $\hat{\lambda}$; 19 to 23 for the $\hat{\alpha}_j$). GRS: $F(20, 1102)$ test that the 20 decile-portfolio α_j are jointly zero. "–": undefined (for the factor itself $\hat{\alpha} = 0$ by construction).